CAM-Guided Multi-Task Learning for Industrial Surface Defect Detection using DenseNet-121

1.1 Dataset Description

The dataset used in this research is Kolektor Surface Defect Dataset 2 (KSDD2), which is a set of grayscale images of industrial production surface images captured at the pixel level. It is very relevant to use this dataset for research on automated inspection of images captured from industrial production surfaces, as it is collected from a controlled environment under real manufacturing conditions [1][4].

The dataset has a total of 3,335 images, out of which 356 images have defects, and 2,979 images are defect-free. The resolution of each image is approximately 230 x 630 pixels. There are various kinds of defects present in the images of this dataset, such as scratches and irregular spots [1].

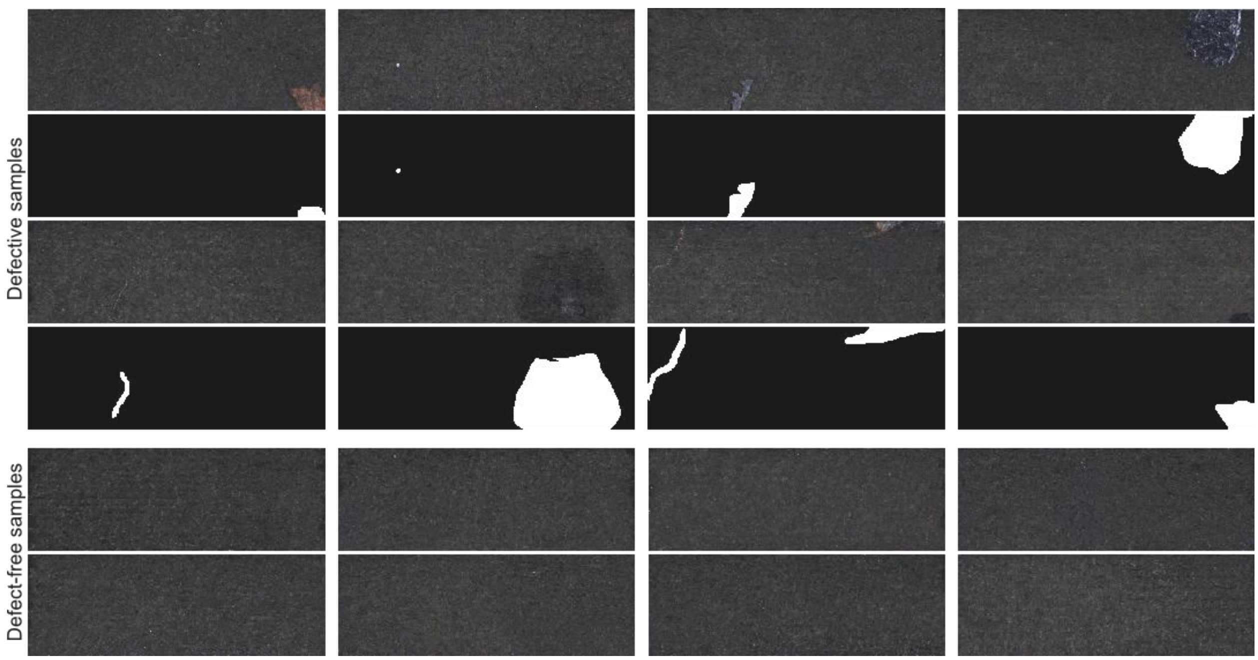
To test the dataset, it has been divided into training and testing datasets. For testing purposes, there are 246 defective images and 2,085 non-defective images in the training dataset, and there are 110 defective images and 894 non-defective images in the testing dataset [1].

Table 1: Dataset Summary Table

Feature	KSDD2
Total Images	3,335
Images with Defects	356
Images without Defects	2,979
Defect Types	Multiple (scratches, minor spots, etc.)
Original Image Size	Approx. 230 x 630 px

Processing Image Size	Not specified, but original size is a good reference
Train/Test Split	Train: 246 positive, 2085 negative Test: 110 positive, 894 negative

Figure 1: Sample Images with Ground Truth Masks



1.2 Data Preprocessing

To ensure stable training and improve generalization performance, various preprocessing operations were conducted on the dataset. First, all images were resized to 256 x 256 pixels to ensure consistent input size and reduce computational costs during training.

Normalization of input data was conducted using mean and standard deviation scaling to help stabilize gradients and accelerate convergence during training. Moreover, data augmentation operations were conducted on the training dataset using Albumentations to increase data variability. Augmentation operations include horizontal flipping and random brightness and contrast changes, simulating various lighting conditions in a real environment.

For classification, labels are generated from segmentation masks, where label 1 is assigned to images with defective pixels and label 0 is assigned to non-defective ones. Note that the dataset is imbalanced in favor of non-defective samples, and strategies such as Dice loss, shared feature learning, and multi-task learning are considered to address this imbalance and improve model generalization

1.3 Model Architecture

The system proposed is based on a CAM-guided multi-task deep learning system, which is intended to carry out classification and segmentation simultaneously. The system is based on the DenseNet-121 model, which is considered to be highly effective in capturing fine-grained patterns on surfaces, as it has dense connections and can propagate gradients efficiently [5]. The weights of ImageNet are also incorporated into this model to ensure better convergence.

To ensure accurate segmentation, a U-Net-based decoder is integrated into this model, which is based on multi-stage upsampling. The model has two heads, one based on classification and another based on segmentation.

Moreover, Class Activation Maps (CAMs) are also integrated into this model to ensure better interpretability, as they are able to provide spatial regions that are significant during classification [6]. This is important to ensure that not only is the system able to provide accurate results, but it is also able to provide explanations.

For training this model, CrossEntropy loss is used for classification, while BCE and Dice loss are used jointly for segmentation. This is important to ensure accurate segmentation and classification.

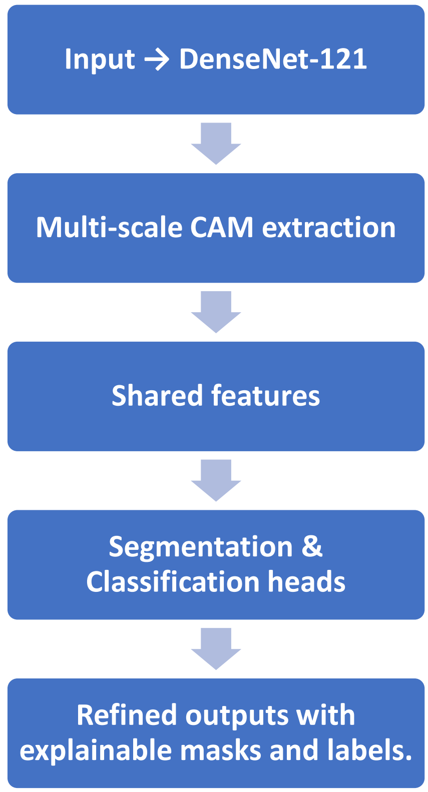
Table 2: Model Parameters

Category	Parameter	Details

Backbone	Encoder	DenseNet-121
	Pretraining	ImageNet pretrained
Architecture	Framework	Multi-task (Classification + Segmentation)
	Decoder	U-Net style (4-stage upsampling)
	Classification Head	Linear (1024 → 2)
	Segmentation Head	Conv (64 → 1)
Loss Function	Classification Loss	CrossEntropyLoss
	Segmentation Loss	BCEWithLogitsLoss + Dice Loss
	Total Loss	CE + 2 × (BCE + Dice)
Optimizer	Type	Adam
Learning Rate		0.0001
Batch Size		8
Epochs		10
Input Size		256 × 256

The encoder of DenseNet architecture provides profound hierarchical features that are applicable to classification and segmentation branches. The decoder increases the resolution of features for accurate placement. The guidance of CAM provides clarity in space with emphasis on areas that influence classification decisions.

Figure 2: Model Architecture Diagram



1.4 Training Setup

The model has been trained with the model configuration set in such a way that the training process occurs in a stable and efficient manner. For instance, the batch size has been set to 8, and the training epochs were set to 10. Moreover, the learning rate has been set to 0.0001, which enables the model to be optimized gradually.

To avoid overfitting, the model has been trained with the addition of callbacks, such as early stopping and checkpointing, during the training process. Moreover, the model has been trained on an NVIDIA GPU, which enables the model to be trained on the local system/Colab environment, thereby making the process smoother. Overall, the training process has been stable, with the model's performance being reliable, as indicated by the constant reduction in the loss.

1.5 Performance Evaluation

The performance of the proposed model was evaluated using appropriate metrics for both classification and segmentation tasks. For the classification task, the model demonstrated strong performance, achieving an accuracy of 95.72% and a precision of 100%, which means the model has been reliable and has not shown any false positive results. For the recall (retrieval) metric, the model has shown 60.91%, which means the model has been unable to detect subtle defect cases.

Overall, the model has shown an F1 score of 75.71%, and the AUROC score of 0.9698 has shown the good discriminative ability of the model for the defect and non-defect classes.

Table 3: Classification Metrics Table here

Classification Metrics	Value
Accuracy	95.72%
Precision	100%
Recall	60.91%
F1 Score	75.71%
AUROC	0.9698

Figure 3: Confusion Matrix

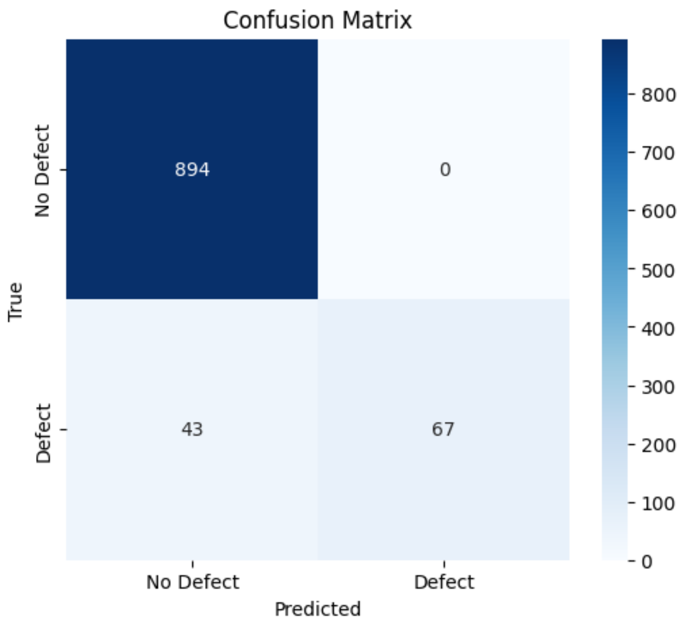
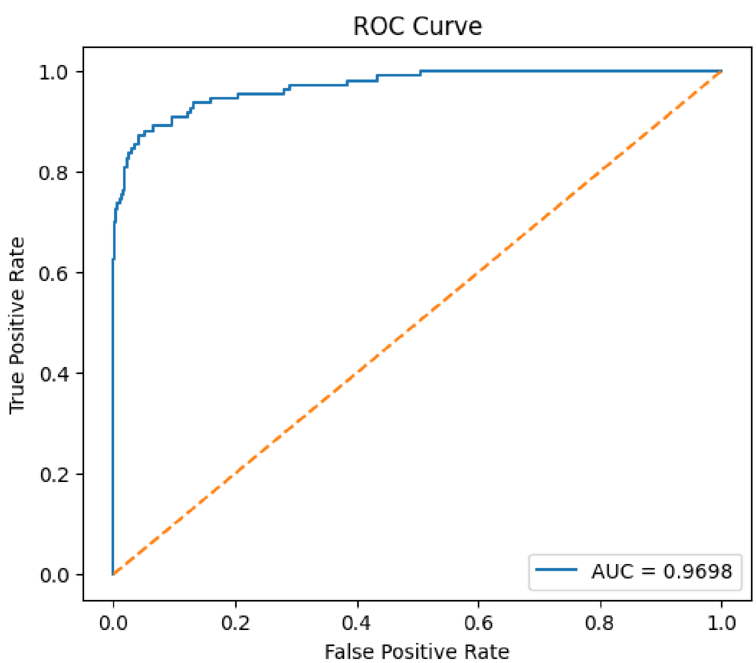


Figure 4: ROC Curve

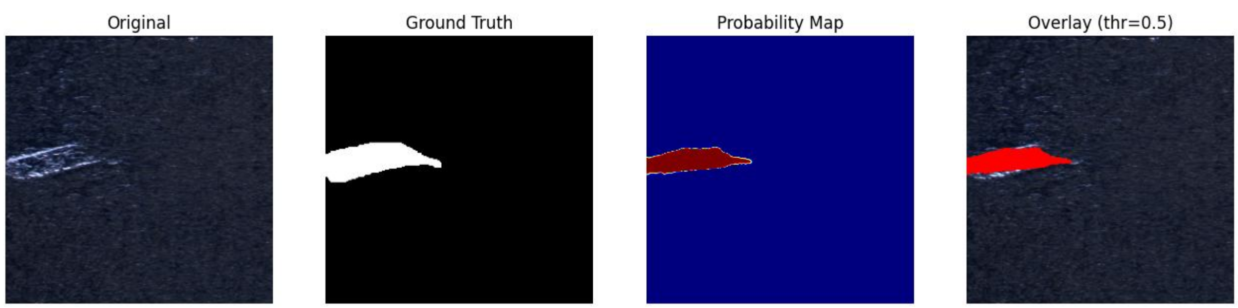


For the purpose of segmentation evaluation, the model presented has shown significant results in terms of localization. The model has been able to obtain a mean Dice metric of 0.9432, which clearly indicates the high degree of overlap between the defect regions and the obtained results. Moreover, the model has been able to obtain a mean Intersection over Union (IoU) of 0.9302, which further indicates the effectiveness of the model in providing precise results for the purpose of segmentation.

Table 4: Segmentation Metrics

Segmentation Metrics	**Value**
Mean Dice Score	0.9432
Mean IoU	0.9302

Figure 5: Segmentation Visualization



1.6 Comparative Benchmarking

The proposed model was compared with existing segmentation benchmarks for the KSDD2 dataset, namely fully supervised U-Net, DeepLabV3+, and RA-CAM with DeepLabV3 [2][3].

These methods represent strong baselines for industrial defect segmentation tasks.

In comparison with existing methods, it is observed that the proposed approach excels in localization and interpretability with the incorporation of CAM. Also, multi-task learning helps in overall performance with shared representations for classification and segmentation tasks.

Table 5: Benchmark Comparison

Model / Method	**Type**	**Key Approach**	**Performance (Reported)**	**Comparison with Proposed Model**
UNet (Fully Supervised)	Segmentation	Encoder–decoder CNN for pixel-level forecasting	58.79% (Defect), 79.40% (mIoU)	Reduced localization accuracy and insufficient classification ability.
DeepLabV3+ (Fully Supervised)	Segmentation	Employs atrous convolution along with the ASPP module, achieving	61.08% (Defect), 80.54% (mIoU)	It serves as a solid baseline but has limitations in multi-task learning and interpretability.
RA-CAM + DeepLabv3 (Weakly Supervised)	Segmentation	CAM-based weak supervision	57.56% (Defect), 78.78% (mIoU)	Competitive yet less precise localization compared to the proposed model.
ExDD (with Synthetic Augmentation)	Anomaly Detection	Dual distribution learning with	97.7% (pixel-wise AUROC)	Emphasizes anomaly detection but lacks

		synthetic data		segmentation precision.
Proposed Model (DenseNet + Multi-Task + CAM)	Classification + Segmentation	Multi-task DenseNet utilizing U-Net decoder and CAM support	**Accuracy: 95.72%, Dice: 0.9432, IoU: 0.9302**	Excellent localization, interpretability, and dependable classification

1.7 Observations & Analysis

Some merits of the proposed model are also identified, such as high segmentation and localization accuracy. Also, the model has achieved zero false positives for classification. This is significant for industrial inspection systems. Additionally, visualization using the CAM approach has improved the model's interpretability by visualizing the spatial regions that are relevant for model prediction [6].

Some limitations identified for the proposed model are moderate recall ability because of conservative decision-making, leading to failure to detect defects. Failures are observed for very small defect regions. As can be inferred from class-wise results, there is a strong performance of non-defective samples and high precision of defective samples.

Figure 6: CAM Visualization (Non-Defect Sample)

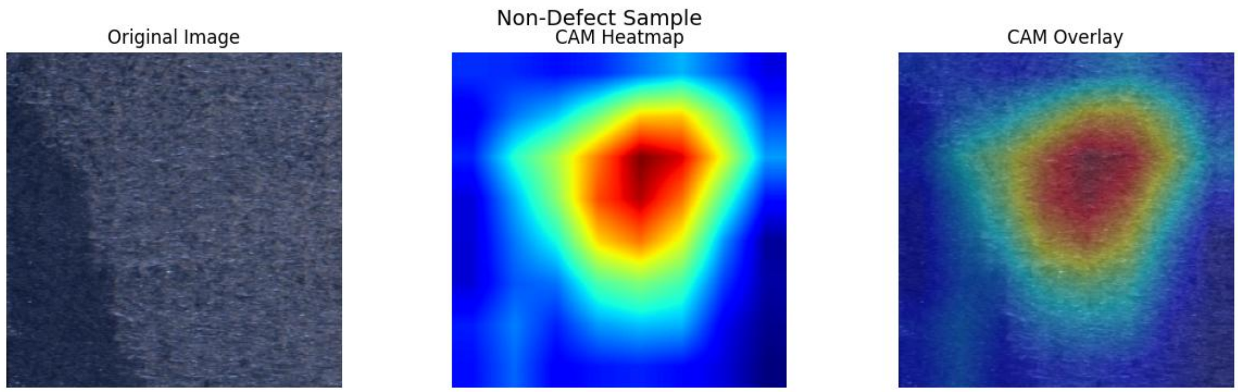
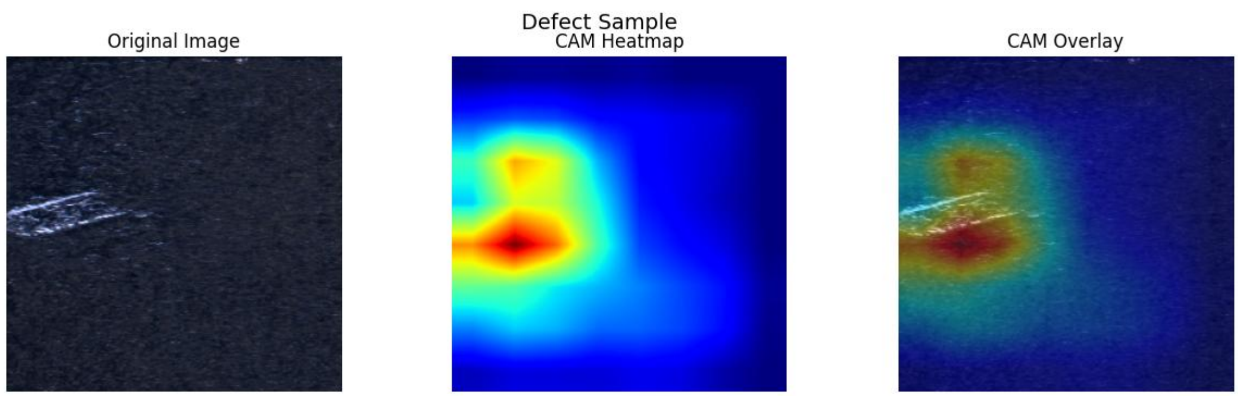


Figure 7: CAM Visualization (Defect Sample)



Figures 6 and 7 show the CAM visualization results of both cases. In defective samples, it is clear that heatmaps are emphasizing the true defect regions, which confirms that the decision-making process is based on meaningful spatial features [6].

The effectiveness of the proposed architecture is due to the strong ability of DenseNet in representing features in decision-making processes [5].

References

1. Y. Xia, J. Lin, et al. "High-Resolution Weakly-Supervised Defect Segmentation on KolektorSDD and KolektorSDD2 Datasets." arXiv preprint, 2025.
2. Y. Xia, J. Lin, et al. "Explainability-guided weakly supervised semantic segmentation for industrial surface defect detection." arXiv preprint, 2025.
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4. Kolektor Group. "Kolektor Surface-Defect Dataset (KolektorSDD and KSDD2)." https://www.vicos.si/resources/kolektorsdd/
5. DenseNet-121: Gao Huang, Zhuang Liu, Laurens van der Maaten, Kilian Q. Weinberger."Densely Connected Convolutional Networks," CVPR 2017.
6. Class Activation Maps (CAMs): Bolei Zhou, Aditya Khosla, Agata Lapedriza, Aude Oliva, Antonio Torralba."Learning Deep Features for Discriminative Localization," CVPR 2016 (published on arXiv in 2015)